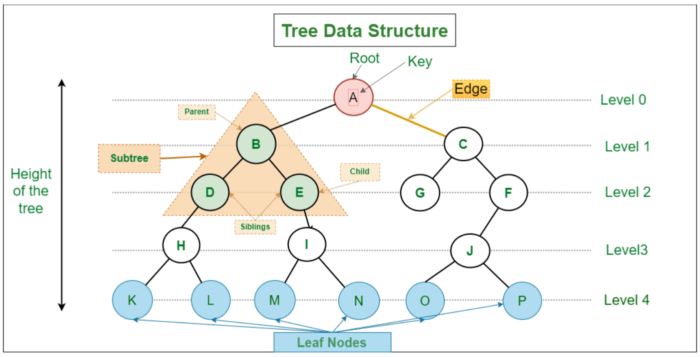
Terms:



1) **Root** - The root, is the ***starting*** position where the first node sits.

The root in the image about is {***A***}.

---

2) **Parent** - To easily explain this, ***Root*** = starting $\longrightarrow \forall n, n =$ node that come from the root, are know as ***children*** and root would be the ***Parent***.

B = { ***D, E*** } where D, E are children and B is the Parent $\land$ ***A*** = { ***B, C*** } is the Parent.

3) **Child** - Again, let's make it easy - above I basically explained it... but if you don't understand... the ***child*** in the case about would be the nodes descending from the ***root***. (see above case)

4) **Sibling** - Those who are ***children*** to the same ***Parent*** are ***Siblings***.

with the ***B*** = { ***D, E*** } case - D, E are siblings! They both have B as a Parent.

---

5) **Descendents** - This is hard to explain without a visual but essentially if you pick a node within a tree - it's desencdants are the nodes that are ***Below*** it.

B = { ***D, E, H, I, K, L, M, N*** } are all descendents of B.

---

6) **Ancestor** - Ancestors are the nodes that are '***above***' the current node being looked at (which I will elaborate more on)

L = { ***H, D, B, A*** } are all nodes that are Ancestors of L.

---

7) **Degree of node** - The degree is hard to explain without a visual - but the ***Degree*** is essentially the amount of

8) **Internal/External (Terminal/Non-Terminal)** - Nodes that leafs are ***Non-Terminals*** $\forall$ other nodes they are said to be ***Terminals***.

9) **Levels** - Levels are the ***depth*** of a tree, starting at 1 index. (Height starts at 0)

10) **Height** - The height of the tree starting from 0 index, essentially when you draw it out you will assume the ***root*** to be the [0] index.

11) **Forest** - essentially a forrest is when you have a collection of trees.

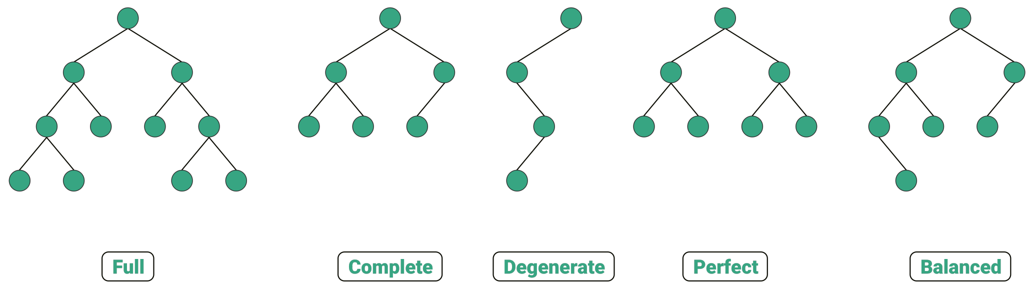


Figure 1: Tree Structures

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1) ***Full*** - A full binary tree is also a ***Strict Binary Tree***!

2) ***Complete*** - A tree that is filled with nodes from ***Left to right*** - or fills by lvl.

3) ***Degenerate*** - A tree that is Degenerate is a tree that

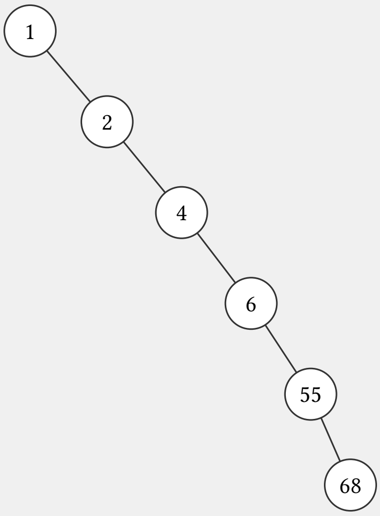
4) ***Perfect*** - A tree that is Perfect $\longrightarrow$ the tree will have the ***Maximum amount of nodes possible*** $\longrightarrow n = 2^{h+1} - 1$

This one is Also a ***Strict Binary Tree*** but also a ***Complete Binary Tree***.

5) ***Balanced*** - A tree that is said to be ***Balanced***

Examples

Example 1:



Is this Complete?

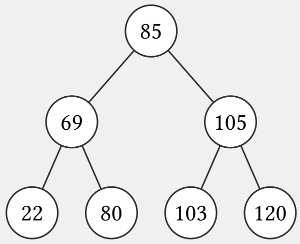
Is this Full?

Is this Perfect?

Is this Balanced?

Is this Degenerate?

Example 2:



Is this Complete?

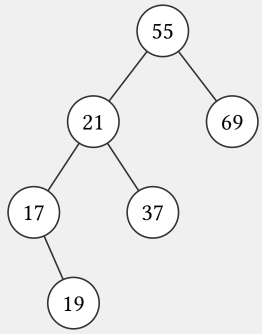
Is this Full?

Is this Perfect?

Is this Balanced?

Is this Degenerate?

Example 3:



Is this Complete?

Is this Full?

Is this Perfect?

Is this Balanced?

Is this Degenerate?